

Lecture Orientation: From Recognition to Response

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- Antigen recognition is only the beginning
- This lecture focuses on events *after* activation
- Expansion, shaping, restraint, and resolution determine outcomes
- Downstream control defines protection versus pathology

Learning Objectives and Conceptual Roadmap

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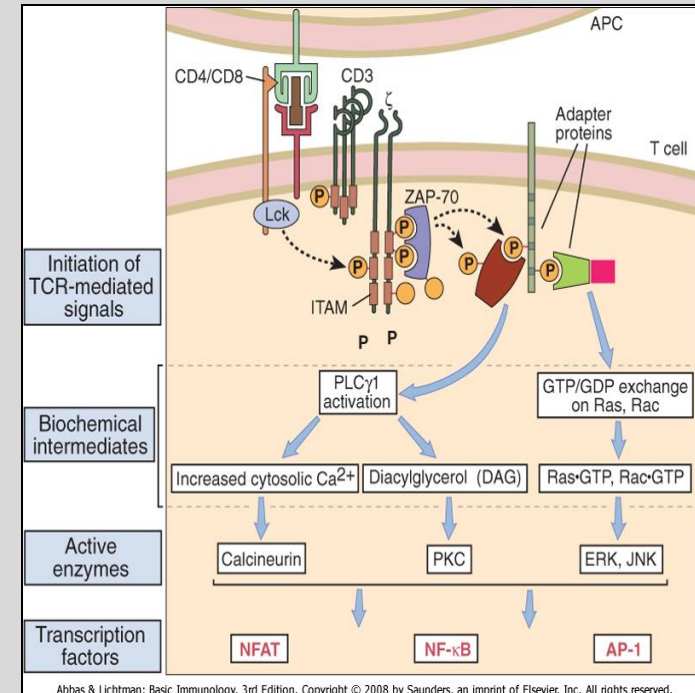
- Understand how activation becomes functional immunity
- Explain expansion, differentiation, memory, and regulation
- Emphasize *why* regulation is necessary
- Core theme: power requires restraint

Why Clonal Expansion Is Essential

- Recognition alone cannot control disease
- Expansion generates sufficient effector numbers
- Amplifies rare specific cells
- Expansion is powerful but dangerous

Signals That Support Expansion

- Antigen recognition is insufficient
- Costimulation promotes survival and proliferation
- Cytokines reinforce expansion decisions
- Context determines whether expansion occurs

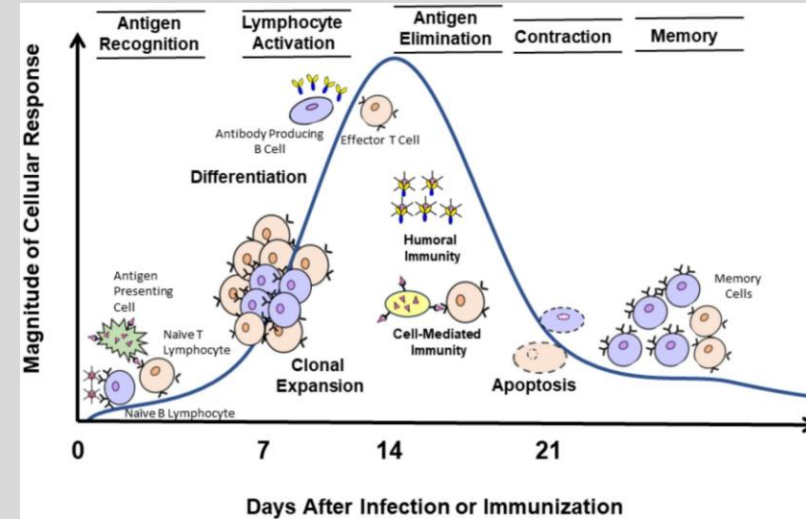


Memory as a Deliberate Immune Strategy

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- Some activated cells become long-lived
- Memory enables faster secondary responses
- Memory formation is intentional
- Defining feature of adaptive immunity



Eph, et al. Composition, Stabilization and delivery of virus-based vaccines: considerations to break free of the cold chain. *Advanced Drug Delivery Reviews*, Vol 227, Dec 2025.

The Need for Controlled Contraction

- Immune responses must end
- Persistent activation causes damage
- Contraction restores balance
- Failure leads to chronic inflammation

Regulation and Tolerance Mechanisms

- Regulation limits immune damage
- Tolerance prevents responses to self
- Essential for immune safety
- Prepares for deeper discussion

Clonal Expansion as a Quantitative Solution

- Antigen-specific cells are initially rare
- Expansion creates large identical populations
- Converts recognition into clinical impact
- Precision without expansion is ineffective

CD8 T-Cell Effector Differentiation

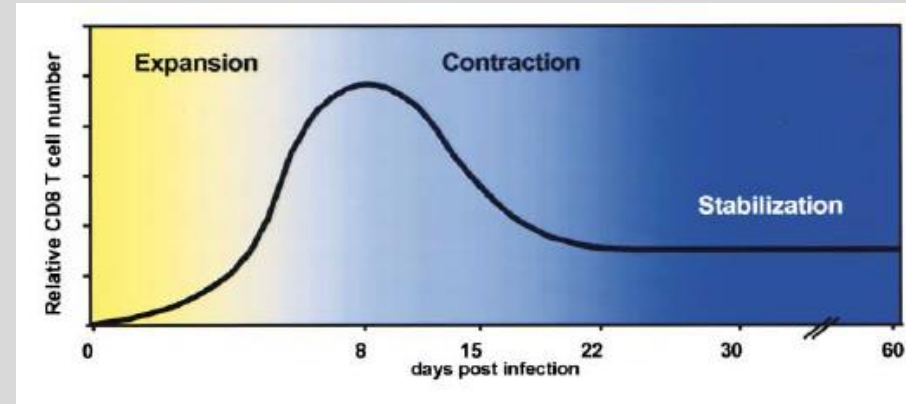
- CD8 cells become cytotoxic effectors
- Acquire killing machinery
- Differentiation is tightly regulated
- Prevents collateral tissue damage

Why Effector Differentiation Must Be Regulated

- Effector programs affect host tissues
- Excess causes damage
- Insufficient responses fail
- Regulation balances effectiveness and safety

Linking Expansion to Immune Function

- Population size matters
- Population composition matters
- Expansion and function are inseparable
- Coordination determines outcome



Hemby, S., et al. Molecular and Functional Profiling of Memory CD8 T Cell Differentiation. *Cell*: 111(6):837-51.

Effector Versus Memory Cell Fates

- Same precursors, different outcomes
- Effector: immediate function
- Memory: rapid recall and persistence
- Dual strategy ensures protection

Homeostatic Maintenance of Memory Pools

- Memory persists without antigen
- Maintained by survival cytokines
- Avoids chronic inflammation
- Ongoing regulation in health

Clinical Implications of Failed Contraction

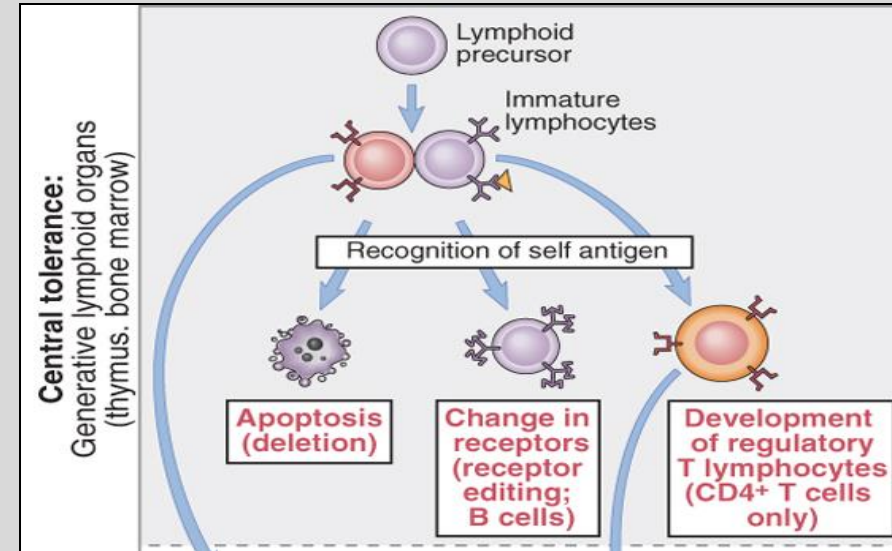
- Persistent effectors drive pathology
- Seen in chronic inflammatory disease
- Therapies aim to restore regulation
- Control loops matter clinically

Transition to Tolerance and Regulation

- Contraction leads into tolerance
- Prevents attack on self
- Failures cause autoimmunity
- Shift to regulation focus

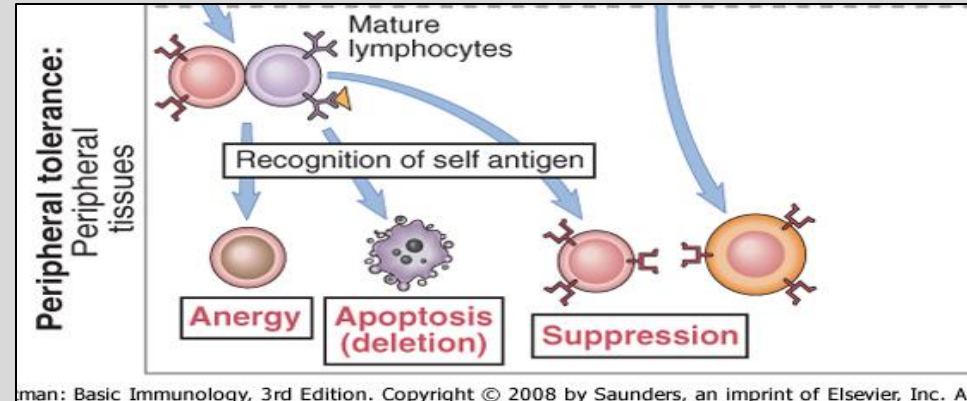
Why Tolerance Is Required

- Receptor specificity risks self-reactivity
- Tolerance controls dangerous clones
- Essential for survival
- Failure underlies autoimmune disease



Central Versus Peripheral Tolerance

- Multiple tolerance layers
- Central: development
- Peripheral: mature lymphocytes
- Redundancy provides safety



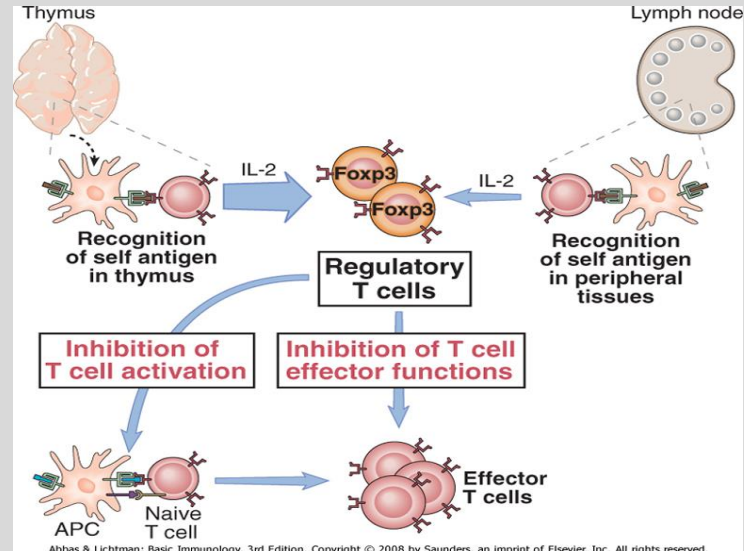
Peripheral Tolerance: An Ongoing Requirement

- Controls mature lymphocytes
- Self antigens encountered lifelong
- Mechanisms include anergy, deletion, regulation
- Dynamic process

FOXP3 and Regulatory T-Cell Identity

- FOXP3 defines regulatory program
- Required for suppressive function
- Loss causes immune dysregulation
- Regulation is active

The **FOXP3** gene provides instructions for producing the forkhead box P3 (**FOXP3**) protein. The FOXP3 protein attaches (binds) to specific regions of DNA & helps control the activity of genes that are involved in regulating the immune system.



Consequences of Failed Regulation

- Autoimmunity and inflammation
- Breakdown of tolerance pathways
- Explains disease mechanisms
- Informs therapy

Integration of Tolerance With Immune Responses

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- Regulation shapes, not silences immunity
- Contextual control
- Balance is central theme
- Sets stage for transplantation

Peripheral B-Cell Tolerance Mechanisms

- Controls escaped self-reactive cells
- Includes anergy and deletion
- Requires T-cell help
- Lifelong control

Why T-Cell Help Is Required for Antibody Responses

- Acts as safety checkpoint
- Prevents autoantibody production
- Integrates tolerance layers
- Loss contributes to disease

Integration of Tolerance and Clinical Practice



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- Central to autoimmunity and transplantation
- Mechanism-based reasoning
- Scales across specialties
- Immunology as framework

Key Concepts

- Activation → expansion → differentiation → regulation
- Tolerance prevents damage
- Memory preserves protection
- Balance is essential

Summary Slide

- **Summary Point 1: Adaptive Immunity Requires Controlled Amplification**

- Antigen recognition alone is insufficient to control infection
- Clonal expansion converts rare antigen-specific lymphocytes into effective populations
- Differentiation assigns specialized effector and helper functions
- Expansion without regulation leads to host tissue damage

- **Summary Point 2: Regulation and Tolerance Are Essential for Immune Safety**

- Contraction actively terminates immune responses after pathogen control
- Memory preserves protection while minimizing ongoing activation
- Central and peripheral tolerance prevent immune responses to self
- Regulatory mechanisms balance protection with long-term tissue integrity

- Core References:

- Abbas AK, Lichtman AH, Pillai S. *Cellular and Molecular Immunology*. Elsevier.
- Murphy K, Weaver C. *Janeway's Immunobiology*. Garland Science.
- Parham P. *The Immune System*. Garland Science.
- Klein J, Sato A. "The HLA system." *New England Journal of Medicine*.